



CYBERSTER

Internship Program - Batch 2

NetSec Internship - Week 1 Task Report

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1. Executive Summary

The process of fetching a webpage appears instantaneous to the end-user, but it relies on a complex, highly orchestrated sequence of events running through the Open Systems Interconnection (OSI) Reference Model. This report analyzes the end-to-end workflow of a client device retrieving a secure webpage via the Google Chrome browser. By dissecting this process through the 7 layers of the OSI model, this document identifies the specific functions occurring at each phase, the inherent security threats present, and the defense-in-depth mitigations required to secure enterprise network communications.

2. High-Level Web Retrieval Workflow

When a user opens Google Chrome and navigates to a secure website (e.g., <https://www.youtube.com>), the following high-level sequence occurs before the page is rendered:

- **URL Parsing & DNS Resolution:** The browser parses the URL and queries a Domain Name System (DNS) server to translate the human-readable domain into a target IP address.
- **Connection Establishment:** The client device initiates a TCP 3-Way Handshake (SYN, SYN-ACK, ACK) with the destination server over Port 443.
- **Cryptographic Handshake:** A TLS (Transport Layer Security) handshake occurs to negotiate cipher suites, verify server certificates, and establish a secure, encrypted tunnel.
- **Data Transmission:** Chrome sends an HTTP GET request through the encrypted tunnel. The server processes the request and sends back the requested HTML, CSS, and JavaScript.
- **Rendering:** The browser decapsulates the incoming data, processes the code, and renders the visual webpage for the user.

3. OSI Model Layer-by-Layer Analysis

To fully understand the security implications of this workflow, the process must be broken down into the 7 layers of the OSI model. Data travels down the layers on the client side (Encapsulation) and up the layers on the server side (Decapsulation).

Layer 7: Application Layer

This layer is the closest to the end-user and interacts directly with software applications that require network communication.

Workflow Action

Google Chrome generates the HTTP GET request and initiates the DNS resolution process to find the server's IP address.

Security Threats

- Phishing & Social Engineering: Malicious sites masquerading as legitimate portals to steal credentials.
- Cross-Site Scripting (XSS) / SQL Injection: Malicious payloads delivered via the web application.
- DNS Spoofing: Redirecting the user's DNS query to a malicious IP.

Mitigation Strategies

Implement Web Application Firewalls (WAF), enforce Content Security Policy (CSP) headers in the browser, and deploy DNSSEC to ensure DNS query authenticity.

Layer 6: Presentation Layer

This layer translates, formats, and most importantly, encrypts/decrypts the data for the Application layer.

Workflow Action

The browser and server perform the TLS handshake here. The data format (like JSON or HTML) is structured, and the payload is encrypted so it cannot be read in transit.

Security Threats

- SSL Stripping: Attackers forcing the browser to downgrade from HTTPS to unencrypted HTTP.
- Weak Cipher Exploitation: Exploiting outdated encryption protocols (like SSLv3 or TLS 1.0).

Mitigation Strategies

Enforce HTTP Strict Transport Security (HSTS) to mandate encryption. Configure web servers to only accept modern protocols (TLS 1.2 and TLS 1.3) and strong cipher suites.

Layer 5: Session Layer

This layer establishes, manages, and terminates the communication sessions between the local and remote applications.

Workflow Action

The browser maintains a persistent session with the web server, often tracking the user's authenticated state via session cookies.

Security Threats

- Session Hijacking: Attackers stealing an active session cookie to impersonate the user.
- Session Fixation: Forcing a user to authenticate using a known session ID.

Mitigation Strategies

Secure session cookies using the HttpOnly and Secure flags. Implement aggressive session timeouts and regenerate session IDs upon user login.

Layer 4: Transport Layer

This layer is responsible for the reliable delivery of data. It segments data, manages flow control, and handles error recovery.

Workflow Action

The TCP 3-way handshake occurs here. The HTTP request is broken down into TCP segments, labeled with source ports (random high port) and destination ports (Port 443 for HTTPS).

Security Threats

- SYN Floods (DDoS): Overwhelming the server with initial connection requests (SYN packets) but never completing the handshake.
- Port Scanning: Attackers scanning the server to find open, vulnerable services.

Mitigation Strategies

Deploy Stateful Firewalls to monitor active connections. Enable SYN Cookies on servers to mitigate SYN flood resource exhaustion.

Layer 3: Network Layer

This layer handles logical addressing and routing data across disparate networks (the Internet).

Workflow Action

The TCP segments are encapsulated into IP Packets. The packet is stamped with the client's Source IP and the server's Destination IP. Routers read this information to forward the packet across the internet backbone.

Security Threats

- IP Spoofing: Attackers altering the source IP address to bypass filtering or execute reflection attacks.
- Ping Floods / Volumetric Attacks: Crashing the network infrastructure with massive amounts of ICMP or IP traffic.

Mitigation Strategies

Implement Access Control Lists (ACLs) on routers to drop spoofed packets (egress/ingress filtering). Utilize Anycast networks and Cloud DDoS protection services.

Layer 2: Data Link Layer

This layer facilitates node-to-node data transfer across a local physical network segment.

Workflow Action

The IP packets are encapsulated into Ethernet or Wi-Fi Frames. The frames are addressed using physical MAC addresses. The local switch uses these MAC addresses to deliver the frame from the PC to the default gateway (router).

Security Threats

- ARP Spoofing/Poisoning: An attacker broadcasting fraudulent ARP messages to intercept traffic meant for the default gateway (Man-in-the-Middle).
- MAC Flooding: Overwhelming a switch's MAC address table to force it to broadcast all traffic.

Mitigation Strategies

Configure Dynamic ARP Inspection (DAI) and Port Security on managed network switches to strictly control MAC address access.

Layer 1: Physical Layer

The lowest layer translates digital data into physical signals (electrical, optical, or radio waves) for transmission over the network media.

Workflow Action

The local Network Interface Card (NIC) converts the frames into electrical pulses (for Ethernet cables) or radio frequencies (for Wi-Fi) and sends them to the local access point or switch.

Security Threats

- Wiretapping/Physical Interception: Physically tapping into copper or fiber lines to duplicate traffic.
- Signal Jamming / Rogue APs: Flooding Wi-Fi frequencies with noise or setting up fake "Evil Twin" Wi-Fi access points.

Mitigation Strategies

Enforce strict physical security for server rooms and wiring closets. Use WPA3 encryption for wireless networks and deploy Wireless Intrusion Detection Systems (WIDS) to monitor for rogue access points.

4. Conclusion

The retrieval of a single webpage demonstrates the vast complexity of modern network communications. Because an attack can occur at any of the 7 OSI layers, from a physical wiretap at Layer 1 to a sophisticated XSS payload at Layer 7, network security cannot rely on a single solution. Organizations must employ a "Defense in Depth" strategy, applying specific, layer-appropriate mitigations to ensure the integrity, confidentiality, and availability of data as it traverses the network.